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ORIGINAL PAPER

The food bank resource allocation problem

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Abstract

One of the global strategic areas in the fight against hunger is the one related to food banks. The mission of food banks is to provide food to people that are in extreme poverty and famine. However, food banks do not have enough resources to supply food to the needy. Hence, hard decisions have to be made every day to determine who will be served, what kind of products, and how many of them will be supplied. In this work, we introduce an optimization model for the Food Bank Resource Allocation Problem, which takes into account inventory management, purchases, product-beneficiary compatibilities, balanced nutrition, and priority of beneficiaries. We also propose an adaptive heuristic to solve large instances of this problem. The mathematical formulation and the proposed heuristic are evaluated over a large set of instances that have been randomly generated based on a real situation of a local food bank. Computational results reveal that our heuristic is able to produce good quality solutions in short computation times.

Keywords Food banks · Humanitarian logistics · Food distribution · Resource allocation

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1 Introduction

Every year, the world population is increasing as well as the deforestation by human hands, which accelerates the erosion of land that could be used for growing food. In addition, the price of food products is so unstable that poor people cannot afford nutritious food. As a result, we have an alarming growth in the number of people who do not have access to adequate and balanced food.

There are several articles in the literature that give figures on malnutrition and food poverty, showing that this problem is really alarming, and although some work has been done on this issue, a greater effort is still pending. According to the Food and Agriculture Organization (FAO) of the United Nations, between 2012 and 2014, at least 805 million people around the world suffered food insufficiency and hunger. Moreover, one in every seven people has a diet that does not provide the necessary daily energy (FAO et al 2012). According to Godfray et al. (2010) this problem represents an increasing challenge, since with population growth the competition for land and resources exploitation also grows, adversely affecting the ability to produce food.

Reducing hunger requires the right combination of several factors, for example, increasing agricultural productivity, improving access to food supplies, and better management of them (Setti et al. 2018; Schanes et al. 2018; Lazzarini et al. 2018; Piergiuseppe et al. 2018). FAO et al (2012) consider also the development of rural areas and implementation of specific nutrition programs. Therefore, we have a great opportunity to apply operations research with the aim of helping the decision maker (food bank manager) to address an efficient allocation of resources. The motivation for this work is helping decision making in a nonprofit organization: in this case, a food bank, whose main decisions consist of determining who will be served, what kind of products, and how many of them will be supplied on a specific day.

Acoording to González-Torre and Coque (2016), a food bank is an organization that receives donated food and in some cases money and redirects them to the needy. Usually, the donation of food comes from different stages of the food chain (producers, processors, retailers, and consumers) to prevent it from becoming food waste; and they can be obtained from multiple sources such as canceled orders, still edible but not marketable products, low demand, damaged products but still fit to eat, wrong order quantities, etc.; see Alexander and Smaje (2008). In contrast, monetary donations are usually carried out to reduce taxes.

Food banks play an important role in the fight against hunger. According to Wilson and Tsoa (2001), in March 2001, 2.4% of people in Canada received emergency assistance from food banks. In March 2002, the Canadian Association of Food Banks reported that its emergency food program had 747,665 users and 305,000 of them were children under the age of 18 years. Moreover, in 2003, 3 million people in Canada were living in food-insecure households which represented 10.2% of Canadian homes, McIntyre (2003). Besides, the End Hunger Report 2013 Australia (2012) shows that 41% of charitable welfare agencies, in Australia, obtained food from food banks once a week and other 41% received donations once or twice a month.

In this work, we introduce the Food Bank Resource Allocation Problem which takes into account heterogeneous products, inventory management, purchases, product–beneficiary compatibilities, balanced nutrition, varied diet, and priority degree of beneficiaries. The decision consists of determining the subset of beneficiaries that will be served as well as the subset of products assigned to each of them such that it provides the maximum score (accumulated priority of the served beneficiaries) subject to some side constraints. It is worth mentioning that each beneficiary can be a single person (child, young, adult), a family, a community, or an organization (hospital, asylum, orphanage, public school, church) which has an ideal demand, in calories, to satisfy its food need and whose vulnerability degree (priority, or degree of need) has been determined by means of a social study.

The social vulnerability index (SVI) described by Flanagan et al. (2011) is one of the most common vulnerability measures and consists of four domains:

- Socioeconomic status: poverty, unemployed, per capita income, education, and health insurance.
- Household composition/disability: age, single-parent household, etc.
- Minority status, language: minorities, race, and ethnicity.
- Housing/transportation: crowding, no vehicle availability, etc.

To calculate the ideal demand of a beneficiary, one can take into account the number of individuals in it, the daily energy requirements, and the features of each one like gender, age, body size, and body composition (Food and Agriculture Organization of the United Nations 2004), the days covered by the donation, and some other special features (McGuire 2011).

The product–beneficiary compatibilities may have different nature, for example, nutritional (infant milk will be wasted if it is donated to an institution that does not have infants), cultural (religious beliefs about some types of animal products), logistical (if the beneficiary has not the appropriate infrastructure to preserve the product), etc.

Among the contributions of this work, we find a joint optimization model to help on the one hand the recovery and redirection of a large number of food products and, on the other hand, humanitarian aid. Both contributions work together: if the products were not used they would be lost along the entire food chain (harvesting, management, marketing, consumption); if there were no products it would hardly be possible to have the social contribution of helping people in food insecurity. Moreover, to the best of our knowledge, there is not another work in the humanitarian logistics literature that studies the resource allocation in food banks by taking into account specific beneficiary features such as product–beneficiary compatibilities, balanced diet, and priority degree in combination with tactical decisions of inventory management and purchases of heterogeneous products. The most related work is the one from Orgut et al. (2016), in which the authors present mathematical models to ensure an equitable distribution of food between counties, they take into account a user-specified upper bound on the absolute difference between each county's "fair share" of donated food and the proportion of total donated food sent to that county.

From the point of view of operations research, we propose a mixed-integer linear formulation and an adaptive heuristic that combines some features of the ALNS (adaptive large neighborhood search) and GRASP (greedy randomized adaptive search procedure). GRASP is a metaheuristic proposed by Feo and Resende (1989) which consists of an initial solution generator and an improvement phase. GRASP has been widely used for solving complex combinatorial problems such as scheduling problems (Aiex et al. 2003; Rajkumar et al. 2011), and vehicle routing problems (Carreto and Baker 2002; Marinakis 2012), among others. Besides, ALNS is a powerful metaheuristic proposed by Ropke and Pisinger (2006), whose main advantage is the adaptive process that guides the search based on the instance at hand.

The organization of this article is as follows. Section 2 provides a brief review of the state of the art. Section 3 presents a general description and a mixed-integer linear formulation of the food bank resource allocation problem (FBRAP). Section 4 describes the proposed adaptive heuristic. Section 5 shows computational results and discussion of the findings of our study. Conclusions follow in Sect. 6.

2 Related work

In the literature, most publications related to food bank problems focus on one of the following areas: location of depots or distribution centers, forecast demand or donations, and transportation of flow from donors to recipients. The common characteristic of these works is that their objectives deal with improving factors such as travel time, equality between beneficiaries, sustainability, efficiency, etc.

Lauren et al. (2014) work on scheduling pickup and delivery routes from donors location to several charitable agencies either directly from the donor or using food delivery points, which serve as links between donors and recipients. They present a set covering model for assigning charitable agencies to each delivery location. Vehicle capacity constraints, frequency of visits to donors, and work schedules are also considered.

A variant of the location-routing problem was introduced by Solak et al. (2012), in which food is delivered from a central warehouse to charitable agencies. The decision is a tactical decision, consisting of assigning charitable agencies to a set of delivery sites, and also scheduling the routes followed by the delivery vehicles. As in Lauren et al. (2014), they deal with a single product. In our studied problem, there are no such tactical decisions, since each beneficiary must go to the food bank to receive the donations. Moreover, we approach beneficiaries not only as charitable agencies, but also as single persons, families, communities, or organizations.

A study related to forecast food bank donations is due to Luther and Lauren (2015). They predict the number of food products that will be picked up from supermarkets over a planning horizon with the aim of avoiding extra costs of inventory and/or transportation. Four different forecast methods have been evaluated and the best forecast has been reported by a multi-layer perceptron neural network.

Orgut et al. (2016) consider a resource allocation problem, which takes into account two important features: (1) a continuous flow of products, which is a key feature to make the difference between a disaster chain and a long-term stable chain; and (2) the real case faced by charitable agencies, where, in most cases the collected resources are less than the total demand of all beneficiaries. Their performance measure is based on equality among beneficiaries.

In this work, we address the resource allocation problem in food banks as a long-term developed chain tackling cases where the supply is unable to meet the demand. One of the main contributions and differences with Orgut et al. (2016) is that we do not consider a homogeneous flow, but a heterogeneous variety of food products.

The World Food Summit World Food Summit (1996) establishes that “Food security exists when all people, at all times, have physical and economic access to sufficient, safe, and nutritious food that meets their dietary needs and food preferences for an active and healthy life” and according to Barrett (2010) food security can be measured by means of four dimensions. The first dimension is the availability of food, which includes not only the amount of food at hand, but also the quality and diversity of it. The second dimension evaluates the access to food, it includes indicators of physical access and the infrastructure that beneficiaries have. The third dimension verifies stability and includes factors that measure the risk of exposure to food insecurity, volatile food prices, etc. The last but not least important dimension is utilization; it evaluates the correct usage and advantages of food products.

Therefore, to take into account, as far as possible, the above four dimensions, in our work we have included some constraints, based on the works of Garille and Gass (2001), Lancaster et al. (2005), and Okubo et al. (2015), to ensure a balanced diet and a proper use of food products (for example infant milk should be assigned to a beneficiary that has infants). Moreover, our work allows prioritizing beneficiaries according to their vulnerability degree.

Another work in a nonprofit organization is the one proposed by Lien et al. (2014) which presents a sequential resource allocation problem with equitable service wihout taking into account heterogeneous products. The authors define service in terms of the ratio of the allocated amount and the observed demand, and the objective function is to maximize the expected minimum fill rate among customers. A similar case can be found in Guo et al (2009).

In Martins et al. (2011), the authors address a food distribution problem in a food bank. The problem consists of assigning donated products to all beneficiaries that demand help without favoring any institution, and the objective is to maximize the global satisfaction of nutrients needed in the organizations, ensuring that the ratio between the donation and the need of each nutrient is similar to all beneficiaries. Our studied problem also takes into account products with similar nutritional features and products that can be donated only to a certain kind of beneficiaries. One can see that the differences with our work are the following: we consider beneficiary priorities, balanced diet, and purchase of products if it is needed.

3 Problem definition

Given a set of beneficiaries N and a set of products M the FBRAP consists of determining the subset of beneficiaries that will be served as well as the subset of products that will be assigned to each of them in order to maximize the total collected score (which is equivalent to sum of the priorities from the served beneficiaries) and guarantees a minimum contribution of calories (kilocalories), balanced nutrition, and compatibility between products and beneficiaries. Each beneficiary $n \in N$ has an ideal demand D_n, in calories, to satisfy its food need and a vulnerability degree (priority, or degree of need) W_n. Additionally, the total content of calories T_m of each product $m \in M$ as well as the number of calories corresponding to carbohydrates C_m, proteins P_m, and fat L_m are known in advance.

Food banks can have different policies to gather and provide resources. The situation that motivates this work is a local food bank in which a beneficiary $n \in N$ is considered served if and only if the total calories assigned reaches a minimum percentage, called α_n, of the total demand D_n, and the donated amount is between the minimum and maximum requirement per nutrient (carbohydrates, proteins, and fat). Moreover, each product is classified into one of eight classes of products (animal product, cereal, sugar, fat, fruit, vegetable, spice, and legume) and each served beneficiary must receive products from at least β different classes of products. If the previous constraints are unmet, the beneficiary is considered unserved.

A mathematical formulation to tackle the Food Bank Resource Allocation Problem (FBRAP) is described as follows.

3.1 Mathematical formulation

Sets

N Beneficiary set
M Product set
K Class of products set

Parameters

D_n	Demand in calories of beneficiary $n \in N$
W_n	Weight (priority) of beneficiary $n \in N$
$\alpha_n \in [0.20, 1.0]$	Minimum percentage of demand to consider that beneficiary n is served
I_m	Inventory of product $m \in M$
E_m	Purchase cost of product $m \in M$, per unit
B	Available budget ($)
T_m	Total calories per unit of product $m \in M$
C_m	Calories from carbohydrates per unit of product $m \in M$
P_m	Calories from proteins per unit of product $m \in M$
L_m	Calories from fat per unit of product $m \in M$
$[\underline{C}, \bar{C}]$	Minimum and maximum proportion of calories from carbohydrates required in a balanced diet
$[\underline{P}, \bar{P}]$	Minimum and maximum proportion of calories from proteins required in a balanced diet
$[\underline{L}, \bar{L}]$	Minimum and maximum proportion of calories from fats required in a balanced diet
β	Minimum number of different classes of products required to serve any beneficiary

$$F_{mk} = \begin{cases} 1 & \text{if product } m \in M \text{ belongs to class } k \in K \\ 0 & \text{otherwise.} \end{cases}$$

Decision variables

x_{nm}: units of product $m \in M$ assigned to compatible beneficiary $n \in N$. Therefore, if product $m \in M$ is not compatible with beneficiary $n \in N$, the assignment x_{nm} is set to zero.

$$y_n = \begin{cases} 1 & \text{if beneficiary } n \in N \text{ is served,} \\ 0 & \text{otherwise.} \end{cases}$$

$$v_{nk} = \begin{cases} 1 & \text{if any product from class } k \in K \text{is assigned to beneficiary} \\ & n \in N, \\ 0 & \text{otherwise.} \end{cases}$$

w_m: purchased units of product $m \in M$.

3.1.1 Optimization model

$$\text{(FBRAP) Maximize } Z = \sum_{n=1}^{|N|} W_n y_n \qquad (1)$$
$$\text{subject to}$$

$$\alpha_n D_n y_n \le \sum_{m=1}^{|M|} T_m x_{nm} \le D_n y_n \qquad n \in N, \qquad (2)$$

$$\underline{C} \sum_{m=1}^{|M|} T_m x_{nm} \le \sum_{m=1}^{|M|} C_m x_{nm} \le \bar{C} \sum_{m=1}^{|M|} T_m x_{nm} \qquad n \in N, \qquad (3)$$

$$\underline{P} \sum_{m=1}^{|M|} T_m x_{nm} \le \sum_{m=1}^{|M|} P_m x_{nm} \le \bar{P} \sum_{m=1}^{|M|} T_m x_{nm} \qquad n \in N, \qquad (4)$$

$$\underline{L} \sum_{m=1}^{|M|} T_m x_{nm} \le \sum_{m=1}^{|M|} L_m x_{nm} \le \bar{L} \sum_{m=1}^{|M|} T_m x_{nm} \qquad n \in N, \qquad (5)$$

$$\sum_{n=1}^{|N|} x_{nm} \le I_m + w_m \qquad m \in M, \qquad (6)$$

$$\sum_{m=1}^{|M|} E_m w_m \le B, \qquad (7)$$

$$v_{nk} \leq \sum_{m=1}^{|M|} F_{mk} x_{nm} \qquad n \in N, k \in K, \tag{8}$$

$$\sum_{k=1}^{|K|} v_{nk} \geq \beta y_n \qquad n \in N, \tag{9}$$

$$x_{nm} \in \mathbb{Z}^+ \qquad n \in N, m \in M, \tag{10}$$

$$y_n \in \{0, 1\} \qquad n \in N, \tag{11}$$

$$v_{nk} \in \{0, 1\} \qquad n \in N, k \in K, \tag{12}$$

$$w_m \in \mathbb{Z}^+ \qquad m \in M. \tag{13}$$

Considering priority as score, the objective function(1) (weighted sum) is to maximize the total collected score coming from the served beneficiaries. Constraints (2) are used to identify which beneficiaries must be served. Constraints (3)–(5) assure that the minimum and maximum requirements of carbohydrates, proteins, and fats are satisfied (balanced nutrition). Constraints (6) establish that the total assignment of products should not exceed the existing inventory and the purchased products. Constraints (7) ensure that the food bank does not spend more than its available budget. Constraints (8) identify which classes of products are assigned to each served beneficiary. Constraints (9) establish the minimum requirement of different classes of products per beneficiary. The remaining constraints impose conditions on the variables.

The minimum and maximum proportions related to each nutrient and the value of β may depend on the beneficiaries characteristics and needs based on the specifications of the Dietary Guidelines for Americans 2010, McGuire (2011). Moreover, if there are no priorities of beneficiaries, $W_n = 1 \, \forall n \in N$.

3.1.2 FBRAP complexity

A particular case of FBRAP arises when there is no available budget ($B = 0$), the minimum number of classes of products required to serve any beneficiary is zero ($\beta = 0$), and the minimum and maximum proportions of Calories related to each nutrient becomes zero and one, respectively (i.e., $\underline{C} = \underline{P} = \underline{L} = 0$ and $\bar{C} = \bar{P} = \bar{L} = 1$). Under such conditions, the simple case reduces to: objective function (1), constraints (2), nature of variables (10), (11) and constraint (6) are reformulated as follows:

$$\sum_{n=1}^{|N|} x_{nm} \leq I_m \qquad m \in M. \tag{14}$$

Therefore, the resulting problem can be seen as a variant of two well-known combinatorial problems: the multiple knapsack problem with assignment restrictions which is NP-hard in the strong sense (Dahl and Foldnes 2006; Dawande et al. 2000) and the subset sum problem also NP-hard (Garey and Johnson 2002).

4 Adaptive heuristic

The proposed solution procedure is based on the main mechanism of the ALNS metaheuristic in which there is a set of destroy and repair operators that attempt to improve a given initial solution (see Algorithm 1). At each iteration, one destroy and one repair operator are chosen with a probability that is proportional to their performance in the past iterations. The process is repeated until the stopping criterion is met and the best-found solution is reported.

In our adaptive heuristic, five destroy operators have been developed and all of them destroy (remove) one or more beneficiaries from the solution at hand. There is only one repair operator, which is also used to generate the initial solution. This repair operator is based on a GRASP and at each iteration, a parameter γ is randomly chosen and used to compute some lower bounds related to the number of calories per nutrient that must be assigned to the selected beneficiaries. The detailed description of our ALNS is given in the following sections.

Algorithm 1: Adaptive Heuristic

```
S_c ← ∅ (Current solution)
Y ← ∅ (Served beneficiaries)
D ← {D_1, D_2, D_3, D_4} (Destroy operators set)
D̄ ← ∅(Forbidden destroy operators set)
T ← ∅ (Forbidden beneficiaries set)
S_c ← RepairOperator(S_c, N, Y, T) (Build an initial solution)
S* ← S_c
for i=0 to 3 do
    Destroy randomly i × 10% of served beneficiaries in S_c        ▷Diversification
    T ← selected beneficiaries that have been destroyed at previous step
    for j=1 to 5 do
        Choose destroy operator δ ∈ D \ D_T
        T ← T ∪ δ(S_c)
        S_c ← RepairOperator(S_c, N, Y, T)
        if S_c is better than S* then
            S* ← S_c
            D̄ ← ∅                                   ▷Set all operators available
            j = 1
        else
            S_c ← S*
            if δ ≠ D4 then
                D̄ ← D̄ ⋃ δ                           ▷Label δ operator as tabu
            end
            j=j+1
        end
        T ← ∅
    end
end
return S*
```

4.1 Construct/repair operator

As we mentioned before, the repair operator and the initial solution generator are the same; therefore, this procedure works from an empty solution or from a partial solution (see Algorithm 2). Let $Y \subset N$ be the set of beneficiaries that are included in the current solution S_c and let $T \subset N \setminus Y$ be the tabu beneficiaries (i.e., beneficiaries whose donations assignment was infeasible or beneficiaries that were previously selected by a destroy operator). Suppose that $H \leq B$ is the amount of money that has been used to buy products. So, $Y = \emptyset$, $T = \emptyset$, and $H = 0$ if $S_c = \emptyset$.

Algorithm 2: Repair Operator

```
Data: S_c, N, Y, T
U ← N \ {Y ∪ T}
repeat
    n̄ ← arg max_{n∈U} W_n / D_n
    Choose γ to compute DC_n̄, DP_n̄, DL_n̄
    (Step 1.) Assign to beneficiary n̄, if possible, one unit of β different products
      which contain the smallest number of Calories and belong to different classes
      of products (AssignmentUpdate(1, m̄, n̄))
    repeat
        Build RCL
        Random selection of product m̄
        Determine the quantity of units (x_n̄m̄) of product m that must be
          assigned to beneficiary n̄ (AssignmentUpdate(x_m̄n̄, m̄, n̄))
    until the assignment of products to beneficiary n̄ has reached the lower
      bounds given by constraints (15)–(17) or RCL = ∅
    Let X_n be the assignments set made to beneficiary n
    if X_n is infeasible then
        Destroy the assignment (UnassignmentUpdate(X_n))
        T ← T ∪ {n̄}
    else
        Y ← Y ∪ {n̄}
        Update U, S_c, etc.
    end
until U = ∅
return S_c
```

At each iteration, the algorithm identifies the unserved beneficiaries and creates the set of candidate beneficiaries $U \subset N \setminus \{Y \bigcup T\}$. Then, the beneficiary $\bar{n}$ with the highest value of the utility function is chosen (i.e., $\bar{n} = \arg\max_{n \in U} W_n/D_n$). After that, a lower bound per nutrient is computed as follows:

$$DC_{\bar{n}} = \frac{\underline{C} + \gamma(\bar{C} - \underline{C})}{(1-\gamma)(\underline{C} + \underline{P} + \underline{L}) + \gamma(\bar{C} + \bar{P} + \bar{L})} \times \alpha_{\bar{n}} \times D_{\bar{n}}, \tag{15}$$

$$DP_{\bar{n}} = \frac{\underline{P} + \gamma(\bar{P} - \underline{P})}{(1-\gamma)(\underline{C} + \underline{P} + \underline{L}) + \gamma(\bar{C} + \bar{P} + \bar{L})} \times \alpha_{\bar{n}} \times D_{\bar{n}}, \tag{16}$$

$$DL_{\bar{n}} = \frac{\underline{L} + \gamma(\bar{L} - \underline{L})}{(1-\gamma)(\underline{C} + \underline{P} + \underline{L}) + \gamma(\bar{C} + \bar{P} + \bar{L})} \times \alpha_{\bar{n}} \times D_{\bar{n}}, \tag{17}$$

where DC_n, DP_n, and DL_n represent the minimum calories from carbohydrates, proteins, and fats, which must be assigned to beneficiary $\bar{n}$ so it can be considered as served. The γ value is randomly chosen from $\Gamma = \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$ and its probability of selection (ρ_γ) is determined by its weight ω_γ which is associated with its past success. The first time the repair operator is called, all $\gamma \in \Gamma$ have the same weight (i.e., $w_\gamma = 1/|\Gamma|$) and in the following steps, the weight and the probability of selection are updated as follows:

$$\omega_\gamma = (S(\gamma) + 1)/(U(\gamma) + 1),$$

$$\rho_\gamma = \omega_\gamma / \sum_{\theta \in \Gamma} \omega_\theta,$$

where $U(\gamma)$ is the number of times γ value has been selected and $S(\gamma)$ is the number of times this value has allowed feasible assignments of food products to beneficiaries. The formula was adapted from Ropke and Pisinger (2006) and it has been used by Salazar-Aguilar et al. (2012).

Once the beneficiary to be served has been chosen and the lower bounds have been computed, a candidate list (CL) of products is created. Each product $m \in$ CL must be compatible with beneficiary $\bar{n}$ and $I_m > 0$ or $B - H \geq E_m$ (i.e., there must be at least one unit of product m in the inventory or enough budget to buy it). Then, our repair operator considers three main steps to carry out the assignment of products to beneficiary $\bar{n}$.

– *Step 1:* Assign to beneficiary $\bar{n}$, if possible, one unit of β different products selected from CL; these products contain the smallest number of calories and contribute with β new classes of products. There are two possible cases, the first case is when $|\text{CL}| < \beta$ or the number of classes from which the products come is less than β. Then, no feasible assignment can be made. The second case is when $|\text{CL}| \geq \beta$ and the number of classes from which the products come is at least β. If the second case occurs, all products in CL are sorted in a decreasing order according to their total calories content and the algorithm selects one unit of β different products that contain the least Calories and contribute, with β classes of products, to the assignment (i.e., select one unit of the β first products that come from β different classes of products). If a feasible assignment can be made go to *Step* 2, otherwise, go to *Step* 4. This step guarantees that the variety of classes of products is satisfied (see constraints 9).
– *Step 2:* Build a restricted candidate list (RCL) from CL, select a product $\bar{m} \in$ RCL, and determine the quantity of units ($x_{\bar{n}\bar{m}}$) that must be assigned to beneficiary $\bar{n}$. In order to build the RCL, the algorithm computes the lower bounds (requirements) per nutrient ($DC_{\bar{n}}, DP_{\bar{n}}, DL_{\bar{n}}$), needed to cover the demand of beneficiary $\bar{n}$, then it identifies the nutrient that is the farthest from its lower

bound, i.e., if indices $\{1, 2, 3\}$ correspond to carbohydrates, proteins, fat and $n_1 = DC_{\bar{n}}$, $n_2 = DP_{\bar{n}}$, $n_3 = DL_{\bar{n}}$, then the nutrient (i^*) with the highest demand to be covered is the one given by $i^* = \underset{i \in \{1,2,3\}}{\operatorname{argmax}} \{n_i\}$. Then, it computes the contribution of each product $m \in$ CL, and only 10% of the products with the best contributions to nutrient i^* are included in the RCL. After that, the products in the RCL are sorted in a decreasing order, taking into account the following function value:

$$\bar{e}_m = \frac{\min\{C_m, DC_{\bar{n}}\} + \min\{P_m, DP_{\bar{n}}\} + \min\{L_m, DL_{\bar{n}}\}}{T_m}. \tag{18}$$

This function measures the contribution rate of calories. See for example the case of carbohydrates: if product m has C_m calories from carbohydrates and the current demand (lower bound) of carbohydrates from beneficiary $\bar{n}$ is $DC_{\bar{n}}$, then, if $C_m > DC_{\bar{n}}$, only $DC_{\bar{n}}$ calories will be taken into account in the evaluation function 18. Finally, a product $\bar{m} \in$ RCL is randomly chosen by using a triangular distribution, i.e., if RCL has cardinality l, the $i - th$ product has a probability $\frac{2 \times (l + 1 - i)}{l \times (l + 1)}$ to be chosen and the quantity of units of product $\bar{m}$ to be assigned to beneficiary $\bar{n}$ is determined as:

$$x_{\bar{m}\bar{n}} = \max\left\{1, \min\left\{I_{\bar{m}} + \lfloor * \rfloor \frac{B - H}{E_{\bar{m}}}, \lfloor * \rfloor \frac{DC_{\bar{n}}}{C_{\bar{m}}}, \lfloor * \rfloor \frac{DP_{\bar{n}}}{P_{\bar{m}}}, \lfloor * \rfloor \frac{DL_{\bar{n}}}{L_{\bar{m}}}\right\}\right\}. \tag{19}$$

Once the assignment of products to beneficiaries ($x_{\bar{m}\bar{n}}$) is determined, the inventory, purchases, and expenses are updated as follows:

- *Step 3:* Repeat *Step 2* until the assignment of products to beneficiary $\bar{n}$ has reached the lower bounds given by (15)–(17) or RCL $= \emptyset$.
- *Step 4:* Let $X_{\bar{n}}$ be the assignment of products made to beneficiary $\bar{n}$.
  - If $X_{\bar{n}}$ is not feasible, then destroy the assignment and return each unit of product to the inventory or dismiss the cost in the current expenses (H), as detailed in Algorithm 4. Then, forbid the selection of beneficiary $\bar{n}$ in the next iterations (i.e., $T = T \bigcup \bar{n}$).
  - Else, add beneficiary $\bar{n}$ to the set of served beneficiaries (i.e., $Y = Y \bigcup \{\bar{n}\}$) and update U, S_c, etc.

The repair operator continues with the next iteration and so on. It stops when there are no more feasible assignments and reports a feasible solution S_c.

After a feasible solution has been created, our adaptive heuristic tries to *diversify the search* by removing a percentage of the served beneficiaries in the current solution. Then, a partial solution is rebuilt with the repair operator.

Algorithm 3: Assignment update.

```
Data: x_{m̄n̄}, m̄, n̄
if x_{m̄n̄} ≥ I_{m̄} then
    I_{m̄} ← 0                                        ▷Units from Inventory
    w_{m̄} ← w_{m̄} + (x_{m̄n̄} − I_{m̄})                 ▷Purchased units
    H ← H + (x_{m̄n̄} − I_{m̄}) × E_{m̄}                  ▷Expenses
else
    I_{m̄} ← I_{m̄} − x_{m̄n̄}                            ▷Units from Inventory
end
DC_{n̄} ← max{DC_{n̄} − x_{m̄n̄} × C_{m̄}, 0}     ▷Update lower bounds per nutrient
DP_{n̄} ← max{DP_{n̄} − x_{m̄n̄} × P_{m̄}, 0}
DL_{n̄} ← max{DL_{n̄} − x_{m̄n̄} × L_{m̄}, 0}
```

Algorithm 4: Unassignment update.

```
Data: X_n
for m=1 to |M| do
    if x_{mn̄} > 0 then
        if w_m ≥ x_{mn̄} then
            w_m ← w_m − x_{mn̄}                        ▷Purchased units
            H ← H − (x_{mn̄} × E_m)                    ▷Expenses
        else
            w_m ← 0                                   ▷Purchased units
            H ← H − (w_m × E_m)                       ▷Expenses
            I_m ← I_m + (x_{mn̄} − w_m)                 ▷Units from Inventory
        end
    end
end
```

4.2 Destroy operators

Five different destroy operators are initially available and all of them except the fifth one have the same probability of being selected. It is important to mention that the fifth operator is used to diversify the search once a local optimum has been found with the other operators.

Then, at each iteration, a destroy operator $\delta \in \{D1, D2, D3, D4\}$ is randomly chosen and its probability of selection is determined by its past performance (same fashion than the selection of γ). It is worth mentioning that the first three operators work in a greedy fashion.

- *D1*. Remove the beneficiary with the highest demand.
- *D2*. Remove the beneficiary with the worst utility value ($R_n = W_n/D_n$).
- *D3*. Remove the beneficiary with the lowest priority.
- *D4*. Remove a random beneficiary.

Once the destroy operator δ has been chosen, the current solution S_c is partially destroyed, according to δ, and the obtained solution is repaired by the repair operator. If the resulting solution is better than S_c, this solution replaces S_c and all operators are set as available. Otherwise, if $\delta \neq D4$, operator δ is set as tabu because it is a greedy operator. Notice that the probability of selection of each available operator is updated at each iteration and the process is repeated until five consecutive non-improvement solutions are generated. If S_c is better than S^* (the best-found solution so far), S^* is replaced by S_c.

Then, the algorithm diversifies the search with the fifth destroy operator (i.e., remove a percentage of served beneficiaries from the current solution S_c) and then reconstructs the resulting solution with the repair operator. After that, the new solution is iteratively destroyed and repaired. As one can see, the diversification of the solution is carried out three times: the first time 10% of the served beneficiaries are removed from the solution at hand, the second time 20%, and the third time 30%. Finally, our algorithm reports the best-found solution S^*.

5 Experimental results

To the best of our knowledge, no previous work has studied the FBRAP and additional constraints that are taken into account in this work. Therefore, we carried out computational experiments to evaluate the FBRAP formulation in several instances. For this purpose, we wrote the models in C++ and solved them by means of CPLEX 12.6 on a 2.10 GHz Intel Xeon(R) CPU E52620 v2 under Ubuntu 14.04 LTS operating system. The stopping criteria per instance were 2 h of computation time and/or a relative gap of 0.01%. Additionally, the proposed adaptive heuristic was coded in C++ and to evaluate its performance, our heuristic was run ten times per instance.

5.1 Instance generator

Intending to generate real-world instances for the FBRAP, we contacted a local food bank which provided us a database containing 200 different products. Then, we calculated the nutrition facts of each product based on the Atwater general factor system (Atwater et al 1906); the energy values are 4 cal/g for carbohydrates and proteins, and 9 cal/g for fat, then the total content for each product is computed as the sum of the calories coming from the three macronutrients. According to the food bank, there are multiple existing units per product and around 80% of the products are compatible with each beneficiary. Therefore, the existing inventory per product was generated as a uniform random value in $[0, 1500] + 1$ and to set the compatibility between any pair (beneficiary, product) a uniform random value in $[0, 1]$ was generated. If the random value is greater than 0.8, then there is compatibility; otherwise, there is no compatibility. Besides, the beneficiary priority has been generated as a value computed as $100 \times (r + 100)$, where r is a uniform random number in range $[0, 200]$ and the demand associated to each beneficiary is a value computed as $70{,}000 \times r_1 + r_2$, where r_1 and r_2 are uniform random values in ranges $[0, 100]$ and $[0, 1000]$, respectively.

Table 1 Characteristics of the instance classes

Class	Beneficiaries	Products
A	100	200
B	300	200
C	500	200

Table 2 Best, average, and worst percentage of relative gap reported by CPLEX

Class	A	B	C
Min	0.05	1.35	1.51
Average	0.94	43.52	38.24
Max	2.40	1045.81	122.50

Varying the number of beneficiaries and products and using the previous specifications of each parameter and different ranges to compute the budget, three instance classes A, B, and C, with 50 instances each, were randomly generated. The budget in classes A, B, and C is a uniform random value in [0, 25,000], [0, 80,000], and [0, 100,000]. See the instance sizes in Table 1.

Additionally, the food bank policies lay down that the minimum number of different classes of products assigned to a served beneficiary must be equal to five, and the minimum and maximum proportion of calories required per nutrient (carbohydrates, proteins, and fats) should be [0.60, 0.70], [0.10, 0.15], and [0.20, 0.30], respectively. Finally, the minimum requirement, α_n, per beneficiary has to be 0.20 of its total demand.

5.2 Results

For validation and evaluation of the formulation proposed for the FBRAP, we tried to solve each instance with CPLEX 12.6, and to reduce the search space and to improve the experimental results, we added the following valid inequality:

$$\sum_{m=1}^{|M|} F_{mk} x_{nm} \leq \bar{M} v_{nk} \quad n \in N, k \in K, \tag{20}$$

where $\bar{M}$ is an upper bound of the amount of products that can be donated to any beneficiary.

In Table 2, we see the best, average, and worst gap reported by CPLEX on each class of instances.

As we can observe, the relative gap reported by CPLEX, for each instance in class A, is very small; however, none of the instances has been solved to optimality. In contrast, CPLEX reported a large relative gap for instances in classes B and C, which in some cases exceeded 100%; this is due to the gap calculation made by CPLEX:

$$\text{Gap}(\%) = 100 \times \left[\frac{\text{BestUpperBound} - \text{BestInteger}}{\text{BestInteger}} \right]. \tag{21}$$

It is important to mention that we tried to solve some instances from class C with a time limit of 24 h; nevertheless, CPLEX still reported high values of gap. Moreover, we have used the solution reported by our heuristic to initialize CPLEX, and after 2 h, CPLEX did not reach a significant improvement.

Finally, we ran our adaptive heuristic ten times and solved each instance in classes A, B, and C. Then, we obtained the best-found solution and the average solution reported by our heuristic over the ten runs. The objective values of these solutions were compared with the best integer solution reported by CPLEX and we computed the relative gap between them.

Figure 1 shows the average computation time (s) per instance (used by each solution method) to reach the best-found solution. CPLEX reached the time limit of 2 h in all instances tested and none of them was solved to optimality. On the other hand, although the average time required by the heuristic to report a solution is growing as the size of the instances increases, the computational time required is only a fraction of that required by CPLEX. The heuristic needs less than 2% of the time used by CPLEX for instances of class A, while for instances of class B and C it uses less than 10 and 17%, respectively.

Table 3 shows that in classes B and C, our heuristic obtained solutions are, on average, at least 20% better than the ones reported by CPLEX. In the last two rows, one can observe the number of instances in which our heuristic and CPLEX obtained better solutions, e.g., for instances in class B, our heuristic outperformed CPLEX in 32 out of 50 instances tested, and for instances in class C, CPLEX reached better solutions only in one case (see Fig. 2).

The relative $\text{Gap}_2(\%)$ reported in Table 3 is computed as:

$$\text{Gap}_2(\%) = 100 \times \left[\frac{Z_C - Z_H}{Z_C} \right], \tag{22}$$

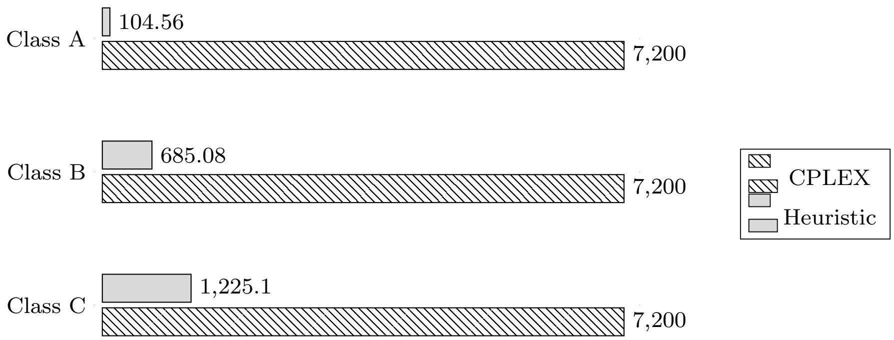

Fig. 1 Average computational time(s) per instance used by each solution method

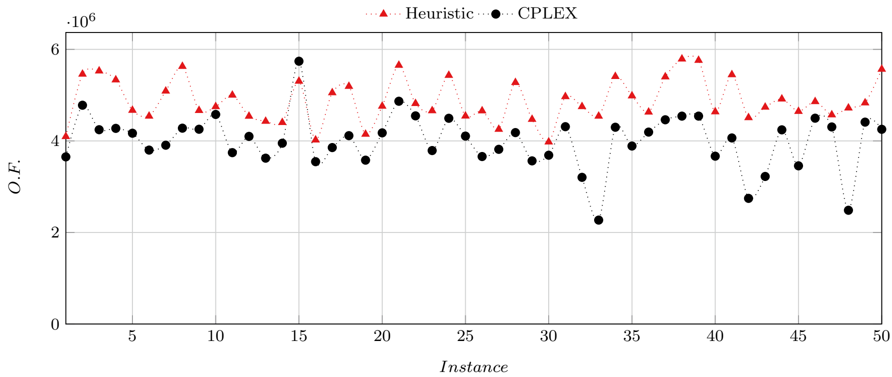

Fig. 2 Objective function value per instance in class C

where Z_C is the best solution reported by CPLEX and Z_H is the best solution found by our adaptive heuristic. A negative gap indicates our heuristic outperformed CPLEX.

Finally, in the final solutions reported by both methods, the amount of food assigned to each selected beneficiary adjusts quite well to the minimum amount of food required per beneficiary, reporting a maximum deviation of 0.23 and 4.37% by our adaptive heuristic and CPLEX, respectively.

5.3 Discussion

Empirical results show that the proposed formulation is sensitive to the combination of nutrition facts of products and the bounds associated with carbohydrates, proteins, and fat. Moreover, this formulation can be used as a support to the decision makers in food banks given that it allows us to solve medium-size instances of the problem with up to 100 beneficiaries and 200 products.

To get an insight into the contribution of the ALNS, the repairing component of the ALNS was replaced by a step that consists of solving the FBRAP with CPLEX, i.e., to carry out the assignment of products we solved the problem by considering only one beneficiary at the time. This version of the ALNS is called hybrid ALNS and the objective value of the best-found solution is named Z_{Hy}; comparative results can be found in Table 4. The third column shows the difference (%) between the objective values reported by CPLEX and the one reported by the hybrid ALNS; the fourth column displays the difference (%) between the objective values found by the ALNS and the hybrid ALNS; finally, the last column shows de Gap(%) between the upper bound ($\bar{Z}_C$) reported by CPLEX and the best-found solution from the hybrid ALNS.

One can see the hybrid ALNS outperformed both, CPLEX and the original version of ALNS. The gap (%) between the best-found solution and the upper bound goes from 2.56 to 7.65% in all tested instances. Therefore, the structure of the proposed ALNS allows us to solve large instances of the problem and to find

Table 3 Average computation time and percentage of relative gap

Class	A	B	C
Time (s)			
Best sol.	104.56	685.08	1225.10
Ave. sol.	93.98	584.98	1089.56
$\text{Gap}_2(\%)$			
Best sol.	16.41	− 21.76	− 24.07
Ave. sol.	18.16	− 20.07	− 22.83
Best solutions			
CPLEX	50/50	18/50	1/50
Heuristic	0/50	32/50	49/50

Table 4 Comparison between the best found solutions reported by different methods

	Gap (%)		
	Z_C vs Z_{Hy}	Z_{H} vs Z_{Hy}	$\bar{Z}_{\text{C}}$ vs Z_{Hy}
Class A			
Average	4.21	− 14.70	5.09
Minimum	1.49	− 22.47	2.56
Maximum	6.64	− 6.34	7.03
Class B			
Average	− 35.37	− 10.98	5.53
Minimum	− 969.97	− 19.16	3.30
Maximum	3.44	− 3.69	7.65
Class C			
Average	− 31.34	− 5.90	4.99
Minimum	− 111.76	− 9.71	3.25
Maximum	3.07	− 2.20	7.17

high-quality solutions. An additional advantage of the ALNS described in Sect. 4 is that any decision -maker at a food bank can use it without needing any software license, and in case he/she has access to a solver, the hybrid version of the ALNS will be the best option to be used.

6 Conclusions

We have introduced the Food Bank Resource Allocation Problem which, allows resource allocation in food banks by taking into account specific beneficiary features such as product–beneficiary compatibilities, balanced diet, and priority degree in combination with tactical decisions of inventory management and purchases of heterogeneous products. Our aim was to present an alternative for the efficient distribution of food in food banks to assure food security and balanced nutrition to people that are food insecure. A mixed-integer linear formulation and an adaptive heuristic were proposed and tested on a large set of instances, which have been generated at random based on a real situation of a local food bank. Computational results revealed that the proposed formulation can be used to solve medium size instances with up to 100 beneficiaries and 200 different products. However, in large instances, our adaptive heuristic outperformed CPLEX with up to 24.07% of relative gap. Moreover, the computation time consumed by our heuristic is significantly shorter than the one required by CPLEX. Besides, if the repairing component of the ALNS is replaced by solving the proposed model with CPLEX, our ALNS is able to find near optimal solutions with a gap (%) from 2.56 to 7.65%.

Additionally, we observed that given the similarities with the knapsack problem, the beneficiaries in the final solution reported by both methods are those that have a high value of the ratio W_n/D_n (priority/demand). Therefore, in practice, a good solution could be obtained by trying to cover the demand of beneficiaries that have the highest values of the mentioned ratio.

The future scope of this research is to make an extension to differentiate and prioritize the products according to their degree of perishability. This is because most of the products that reach a food bank come from commercial chains and most of them have a very short period of time in which they are suitable for consumption.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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